

Accepted Manuscript

Short communication

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PII: S0021-9290(18)30088-5

DOI: <https://doi.org/10.1016/j.jbiomech.2018.02.003>

Reference: BM 8565

To appear in: *Journal of Biomechanics*

Accepted Date: 1 February 2018



Please cite this article as: C.M. Brahms, Y. Zhao, D. Gerhard, J.M. Barden, Stride length determination during overground running using a single foot-mounted inertial measurement unit, *Journal of Biomechanics* (2018), doi: <https://doi.org/10.1016/j.jbiomech.2018.02.003>

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Stride length determination during overground running using a single foot-mounted inertial measurement unit.

Article Type: Short Communication

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Keywords: Stride length, IMU, Sensor, Running, Pedestrian-dead reckoning, ZUPT

Abstract

From a research perspective, detailed knowledge about stride length (SL) is important for coaches, clinicians and researchers because together with stride rate it determines the speed of locomotion. Moreover, individual SL vectors represent the integrated output of different biomechanical determinants and as such provide valuable insight into the control of running gait. In recent years, several studies have tried to estimate SL using body-mounted inertial measurement units (IMUs) and have reported promising results. However, many studies have used systems based on multiple sensors or have only focused on estimating SL for walking. Here we test the concurrent validity of a single foot-mounted, 9-degree of freedom IMU to estimate SL for running. We employed a running-specific, Kalman filter based zero-velocity update (ZUPT) algorithm to calculate individual SL vectors with the IMU and compared the results to SLs that were simultaneously recorded by a 6-camera 3D motion capture system. The results showed that the analytical procedures were able to successfully identify all strides that were recorded by the camera system and that excellent levels of absolute agreement ($ICC(3,1) = 0.955$) existed between the two methods. The findings demonstrate that individual SL vectors can be accurately estimated with a single foot-mounted IMU when running in a controlled laboratory setting.

1. Introduction

The basic spatiotemporal parameters of locomotion, stride length (SL) and stride rate (SR), are important because they dictate movement speed and represent the integrated output of various biomechanical determinants (e.g., muscle strength, limb length, joint range of motion, etc.). Given that speed can be regulated by changing SL or SR, the measurement of one (but not both) of these parameters can result in an incomplete understanding of the interplay between these two variables in relation to locomotor control (Hay, 2002).

While the measurement of temporal parameters such as cadence and SR using inertial-based systems is relatively straightforward (Kobsar et al., 2014; Moe-Nilssen & Helbostad, 2004), SL is typically measured using other methods such as camera systems, force plates or pressure mats, either by recording a few strides in the laboratory or by using a treadmill. Consequently, studies that have investigated the stride characteristics of prolonged overground locomotion have focused on the temporal aspects of the stride pattern (e.g., stride time) or made indirect estimates of the spatial dimension. However, in order to conduct more complex time-series analyses (e.g., fractal analysis) of spatiotemporal parameters during continuous locomotion, accurate information concerning SL is necessary.

Several studies have tried to estimate SL with body-mounted sensors that use beacon technology (i.e., transmitted data) from Global Positioning System (GPS), ultrasound or radio signal systems (Beauregard, 2006; Bichler et al., 2012; Maki et al., 2012). Inertial measurement units (IMUs) offer the advantage that they can operate without pre-installed infrastructure. By using a technique called Pedestrian Dead Reckoning (PDR), IMUs can solve the problem of localization and object tracking by calculating displacement and orientation vectors via the integration of triaxial acceleration and angular velocity information (Feliz et al., 2009; Jimenez et al., 2009; Kim, Jang, Hwang, & Park, 2004). While SL can be determined in theory, its

practical application is complicated due to measurement errors and accumulated noise. Since accurate estimates of the velocity, position and orientation vectors depend on exact knowledge of the initial values, noise can induce substantial drift over time, which is exacerbated during the integration process (Fischer et al., 2013).

Drift error can be greatly reduced when zero-velocity updates (ZUPT) are periodically applied during the integration process (Angermann et al., 2010; Bichler et al., 2012; Feliz et al., 2009; Jimenez et al., 2009). These updates require accurate knowledge about when the velocity of the sensor is at zero so that the integrated velocity can be reset whenever the IMU is stationary, which corrects for the accumulation of noise-produced error. In locomotion, ZUPTs have been successfully applied during the stance phase of each gait cycle when an IMU is mounted on the foot. While this approach has been shown to greatly improve the accuracy of algorithms aimed at object tracking and SL estimation (Feliz et al., 2009), to our knowledge no study has explicitly investigated the accuracy of individual sensor-based SL estimates for overground running. Consequently, the purpose of this study was to determine the concurrent validity of a running-specific, sensor-based ZUPT-algorithm using a 3D-camera motion capture system as the criterion. It was hypothesized that the sensor-based method would be capable of producing excellent levels of agreement with the criterion.

2. Methods

2.1 Participants

Eleven healthy young adults (4 female, 7 male) volunteered to participate in the study. The participants had no medical or pre-existing health conditions and a range of anthropometric characteristics (height, weight, leg length; Table 1) were selected. The study was approved by

the University of Regina Research Ethics Board (REB) and all participants gave written informed consent in accordance with the Declaration of Helsinki.

2.2 Instrumentation and Test Procedures

A 9-degree of freedom IMU (Xsens, MTw) was secured and oriented longitudinally along the right foot of each participant so that anteroposterior, mediolateral and vertical motion were sampled at 100 Hz and detected along the x, y and z-axes, respectively.

Motion of the IMU prior to the experiment was prevented given that Kalman filters require a short period of time to stabilize (i.e., settling time). Three reflective markers (at the heel, the second metatarsal head and directly on the IMU in the area of the mid-foot) were attached to each participant's right shoe to track its position with the camera system during the trials. The front and back shoe markers provided redundant information about foot position that was used only if the mid-foot marker was obstructed or detached during the trial. All marker positions were recorded at 100 Hz. Each participant performed twenty 10 m trial runs through the 3D-camera space, in which a range of typical distance running speeds were tested. For each trial, 1-3 full strides were recorded and used for analysis. To identify the beginning and end of each trial and to ensure a stable gait pattern throughout the trial, participants were asked to stand still outside the calibrated camera volume for 5 seconds and to initiate each run with their left foot. This ensured that the first footfall recorded by both the camera system and the foot-mounted IMU always occurred with the right foot, which allowed for synchronization of the strides for further analysis.

2.3 Data Analysis

Custom MATLAB scripts (v. R2012a, The MathWorks, Natick, MA), based on a modified version of the algorithm proposed by Angermann et al. (2010) and Fischer et al. (2013), were used to analyze the recorded IMU signals. This method uses inertial integration with zero-velocity updates (ZUPT) during stance, in combination with Kalman filtering, to estimate individual SLs during locomotion. Specifically, the method used in this study consisted of the following steps: 1) filter parameters (gravity, sampling rate, accelerometer/gyroscope noise, stance thresholds, ZUPT measurement noise covariance matrix) were defined and the variables (e.g. accelerations, sensor orientation) were initialized; 2) the initial orientation of the sensor was estimated by gyroscopic integration and the accelerations were transformed into a global reference frame and double-integrated to obtain displacement; 3) a zero-velocity test based on static gyroscope and acceleration thresholds was applied to each data frame, to check if the conditions for stance were met. Stance was declared using experimentally determined thresholds of < 3.5 rad/s and $0.3\text{--}1.7$ g for simultaneous resultant angular velocity and acceleration, respectively, and differed from those used in previous studies; 4) when stance was declared, a Kalman Filter estimated velocity, position and orientation errors and corrected the position output. Unlike the methods used by Angermann et al. (2010) and Fischer et al. (2013), an additional step was added to the algorithm, which estimated individual SL vectors by calculating the horizontal distance covered between subsequent stance periods.

Marker coordinate positions were processed using Nexus software (v.1.8, Vicon Motion System, UK). SL was defined as the horizontal displacement of the mid-foot marker at foot contact between subsequent footfalls. Marker trajectories were low-pass filtered using a fourth-order Butterworth filter with a cutoff frequency of 10 Hz. For the camera-based method, stance was defined as occurring whenever the horizontal velocity of the mid-foot marker dropped below a threshold of 0.2 m/s. Stance phase position data were averaged to obtain a single value, and

after the camera and sensor-based SLs were determined the strides were manually matched for comparison.

To establish the accuracy of the IMU-based SLs, the absolute level of agreement between methods was assessed using a model 3 (3,1) intraclass correlation (ICC). This ICC is designed for two-way mixed single measures and quantifies the agreement between two methods on a particular dependent variable. ICCs were classified as poor (< 0.4), moderate ($0.4-0.75$) or excellent (> 0.75) (Shrout and Fleiss, 1979). The Pearson product-moment correlation coefficient was also calculated to quantify the strength and direction of the linear relationship between the two SL estimates. To further compare the two SL methods, the between-method differences were plotted against the averages of the two techniques, using a Bland-Altman plot.

3. Results

A total of 162 trials and 331 strides were recorded by the camera system and included in the analysis. The average running speed was 3.55 m/s (SD = 0.34 m/s), and ranged from 2.71 to 4.36 m/s. The IMU-based SL algorithm successfully identified all of the strides that were recorded by the camera system. An excellent correlation was found between the 3D motion capture and IMU-based methods for the measurement of SL ($r = 0.961$, $\eta = 331$, $p < 0.001$; see Fig. 1). IMU-based mean SLs and standard deviations (2.592 ± 0.276) were similar to those recorded by the camera-based system (2.623 ± 0.272). The mean absolute error of the sensor-based method was 0.05 m (2.01%) compared to the motion capture criterion. The root mean squared error of the two methods was 0.083 m and the ICC revealed excellent levels of absolute agreement (ICC(3,1) = 0.955, $df = 330$, $p < 0.001$). The upper and lower bounds of the 95% confidence interval were 0.927 and 0.970, respectively.

The mean between-method difference in SL estimation was -0.032 m, which suggests a small bias of the sensor-based algorithm to underestimate SL (Fig. 2). The 95% limit of agreement for the between-method difference was 0.118 m and -0.183 m, respectively. A linear regression with SL difference as the dependent variable and mean SL as the independent variable further corroborated the findings that no significant trends existed in the between-method differences for different SLs ($p = 0.350$). Consequently, the null hypothesis that there was no proportional bias in mean SL, was accepted.

4. Discussion

The high ICC (0.955) and correlation coefficient ($r = 0.961$) results demonstrate that the proposed ZUPT-algorithm accurately estimated SL at typical distance running speeds. The current study employed a stance detection algorithm based on static gyroscope and acceleration thresholds. It has been shown that position estimates and SL vectors are largely dependent on the correct identification of the stance phase because ZUPTs are applied at each of these points (Zhao, Brahms, Gerhard, & Barden, 2015). An approach that considers both the rotation rates and the acceleration readings to detect stance has been shown to yield better results over a range of walking and running speeds than approaches based on accelerometer or gyroscope readings alone (Angermann et al., 2010; Fischer et al., 2013). The thresholds were experimentally determined to establish values that worked best for all participants. Additionally, a minimum SL constraint ($SL \geq 0.75$ m) was added to correct for potential insertion errors (Angermann et al., 2010). For SLs below this threshold, the last ZUPT would be declared as invalid and the algorithm would proceed with the integration process until the next ZUPT was applied. As a result, no stance detection errors occurred within the 331 strides that were included in the analysis.

Despite the excellent results, it was noted that some slight variations occurred in the between-method errors across participants (Fig.3). Although we could not investigate the cause of these differences, it is possible that individual variations of technique influenced the signal characteristics, which in turn, affected the accuracy of the SL algorithm. Threshold-based approaches can be sensitive to differences in gait patterns, running surfaces, sensor placement and running speed (Angermann et al., 2010; Bichler et al., 2012) and work best when the foot is relatively stationary during stance compared to when it is moving during contact with the ground (e.g., pronation/supination, plantarflexion/dorsiflexion). While the findings of this study suggest that the IMU-based ZUPT method has practical applications for assessing the SL of overground running outside of the lab, it remains to be seen whether these estimates are more accurate for runners who exhibit a highly stable stance phase versus runners whose feet exhibit more rotational motion during this phase.

5. Conclusion

This study demonstrates that individual SLs can be accurately determined using an IMU-based, PDR/ZUPT approach at representative distance running speeds. This method has potential for real-time monitoring of SL during unconstrained, continuous overground running and should allow researchers to apply more complex time-series analyses (e.g., fractal analysis) to basic locomotor parameters. It provides an accurate and cost-effective alternative to restrictive laboratory methods, as well as the opportunity to monitor SL for the purpose of optimizing performance and investigating factors related to running injuries (e.g., fatigue-related decreases in SL).

Conflict of interest statement

There are no issues of conflict of interest with regard to this manuscript.

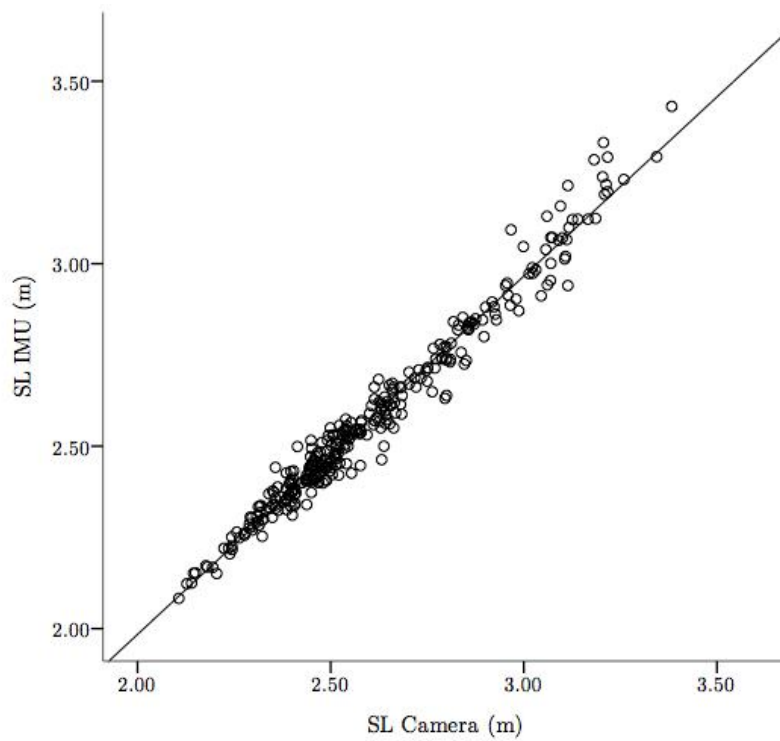
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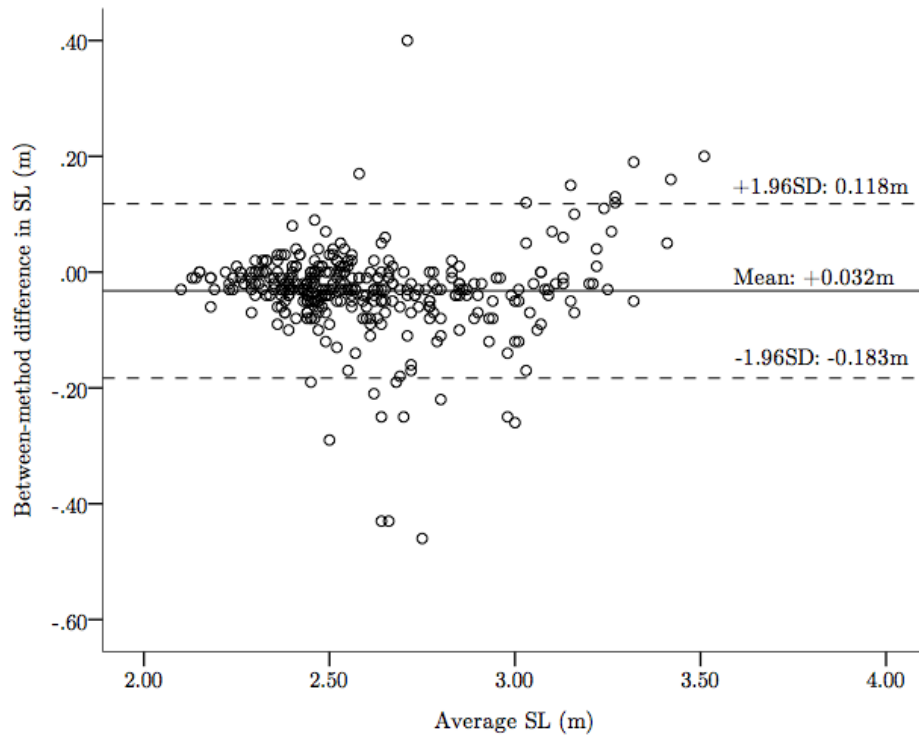
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Figure 1:



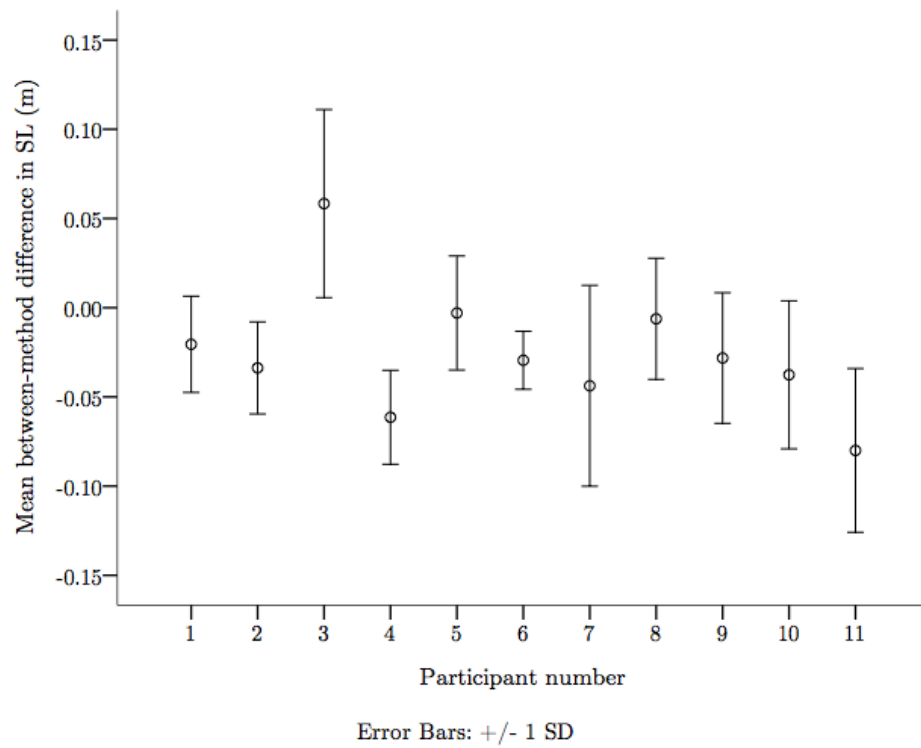
Scatterplot showing the relationship between the stride length estimates of the two different methods.

Figure 2:



Bland-Altman plot comparing the two different methods for calculating stride length.

Figure 3:



Between-method differences in stride length for each participant.

Table 1

Variable	Mean $\pm$ SD
Age (y)	22.3 ($\pm$ 1.5)
Height (cm)	175.2 ($\pm$ 23.1)
Leg Length (cm)	80.56 ($\pm$ 4.36)
Body mass (kg)	76.04 ($\pm$ 3.19)

Characteristics of subjects participating in the study